

Linear Algebra (E)

2025 Fall

Midterm Exam

2025.11.16

Exam Duration: 3 Hours

Name: _____

Student ID: _____

This exam includes 15 pages (including this page) and 5 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam allows one SINGLE sheet (size A4, double-sided is fine). No calculator or any electronic device is allowed. Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 50. If your total score exceeds 50 (there are 53 points in total), it will be recorded as 50.

Problem	Points	Score
1	12	
2	10	
3	13	
4	8	
5	10	
Total:	53	

1. Let A be a 3×4 matrix, and we only know that its RREF is

$$R = \text{RREF}(A) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Write $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4]$ as its columns.

- (a) (3 points) Find all $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{a}_4$. Describe the solution set explicitly.

- (b) (3 points) Is the matrix $[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a}_4]$ invertible? If yes, provide a proof. If not, find its RREF.

- (c) (2 points) Compute $\text{rank}(AR^T)$ and justify your answer.

- (d) (4 points) Suppose that starting from A , we performed the following row operations to get to its RREF.

1. Add twice of row one to row two, and three times of row one to row three.
2. Add row two to row three.
3. Divide row i by i for $i = 1, 2, 3$.
4. Subtract row three from row two.

Find the LDU decomposition of $B = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$, then find B^{-1} .

Answer:

(a) Solve $A\mathbf{x} = \mathbf{a}_4$ by row-reducing $[A \mid \mathbf{a}_4]$:

Since $\text{RREF}(A) = R$ and row operations preserve column relations, the RREF of the augmented matrix is

$$[A \mid \mathbf{a}_4] \xrightarrow{\text{row ops}} [R \mid \mathbf{r}] = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 3 \end{array} \right].$$

Note that the last two columns of $[A \mid \mathbf{a}_4]$ are the same, hence the last two columns of $[R \mid \mathbf{r}]$ must be the same as well.

Thus the reduced system $R\mathbf{x} = \mathbf{r}$ gives

$$x_1 + 2x_4 = 2, \quad x_2 = 0, \quad x_3 + 3x_4 = 3.$$

Let $t := x_4$ be free. Then $x_1 = 2 - 2t$, $x_2 = 0$, $x_3 = 3 - 3t$, $x_4 = t$.

Solution set (set notation):

$$\left\{ \begin{bmatrix} 2 - 2t \\ 0 \\ 3 - 3t \\ t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

(b) Invertibility of $[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a}_4]$ and its RREF:

From the last column of R we read the column relation $\mathbf{a}_4 = 2\mathbf{a}_1 + 3\mathbf{a}_3$, so the 3×3 matrix $[\mathbf{a}_1 \quad \mathbf{a}_3 \quad \mathbf{a}_4]$ is **not** invertible.

By reading the RREF, $\mathbf{a}_1, \mathbf{a}_3$ are independent, and $\mathbf{a}_4 = 2\mathbf{a}_1 + 3\mathbf{a}_3$. So

$$\text{RREF}([\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_4]) = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

(c) Compute $\text{rank}(AR^T)$:

We know $A = ER$ with E invertible, so $AR^T = ERR^T$. We only care about the rank, so the invertible matrix E has no influence. With $R = [I_3 \mid \mathbf{c}]$ and

$$\mathbf{c} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}, \text{ we have}$$

$$RR^T = I_3 + \mathbf{c}\mathbf{c}^T = \begin{bmatrix} 5 & 0 & 6 \\ 0 & 1 & 0 \\ 6 & 0 & 10 \end{bmatrix}.$$

This matrix is “hidden block diagonal”: the middle row/column forms a 1×1 block $[1]$ (invertible), and the remaining 2×2 block $\begin{bmatrix} 5 & 6 \\ 6 & 10 \end{bmatrix}$ is invertible because its two rows $(5, 6)$ and $(6, 10)$ are not parallel. Hence RR^T is invertible and

$$\text{rank}(AR^T) = 3.$$

(d) LU (equivalently LDU) of $B = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3]$, then B^{-1} :

Given forward elimination

$$R_2 \rightarrow R_2 + 2R_1, \quad R_3 \rightarrow R_3 + 3R_1, \quad R_3 \rightarrow R_3 + R_2,$$

we can *read off* the unit-lower factor

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & -1 & 1 \end{bmatrix},$$

and from the scaling step $R_i \rightarrow \frac{1}{i}R_i$ ($i = 1, 2, 3$) we read

$$D = \text{diag}(1, 2, 3).$$

The reverse of the back-substitution $R_2 \rightarrow R_2 - R_3$ is $R_2 \rightarrow R_2 + R_3$; applying this to the identity yields the unit-upper factor

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus an **LDU** decomposition is

$$B = L D U.$$

For the inverse:

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix}, \quad D^{-1} = \text{diag}\left(1, \frac{1}{2}, \frac{1}{3}\right), \quad U^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence

$$B^{-1} = U^{-1} D^{-1} L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{2}{3} & \frac{1}{6} & -\frac{1}{3} \\ \frac{5}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}.$$

2. When A is not invertible, or not even square, how can we still find some sort of “inverse” of A ? The mathematicians Moore and Penrose independently came up with the same idea, called a pseudo-inverse. This is widely used in practice.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \\ 0 & 1 \end{bmatrix}$$

be a 3×2 matrix of full column rank. A 2×3 matrix B is called a *pseudoinverse* of A if

$$ABA = A, \quad \text{and both } AB, BA \text{ are symmetric.}$$

- (a) (3 points) For each of the following products, state whether the multiplication is *defined*, and if so, give its resulting size.

$$A^T A, \quad A^T B, \quad BAB, \quad A^2, \quad \begin{bmatrix} O & A \\ B & O \end{bmatrix} \begin{bmatrix} O & B \\ A & O \end{bmatrix}, \quad \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}.$$

- (b) (3 points) Compute $A^T A$ and AA^T . Find their ranks and state which one is invertible. If invertible, compute its inverse explicitly.

- (c) (2 points) Prove the following formula

$$B = (A^T A)^{-1} A^T.$$

Then compute the entries of B . (Hint: Start with $ABA = A$, use the fact that AB is symmetric.)

- (d) (2 points) For a polynomial $p(x) = x^4 - 3x^2 + 2x + 1$, calculate $p(AB)$.

Answer:

1.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}, \quad B \in \mathbb{R}^{2 \times 3}.$$

- $A^T A$: defined, size 2×2 .
- $A^T B$: not defined (2×3 times 2×3).
- BAB : defined, size 2×3 .
- A^2 : not defined (3×2 times 3×2).
- $\begin{bmatrix} O & A \\ B & O \end{bmatrix} \begin{bmatrix} O & B \\ A & O \end{bmatrix}$: defined; each factor is 5×5 , so the product is also 5×5 .
- $\begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$: not defined because we cannot even build block matrices as these.... A, B have different number of rows, and different number of columns.

2.

$$A^T A = \begin{bmatrix} 5 & 0 \\ 0 & 6 \end{bmatrix}, \quad AA^T = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 5 & -1 \\ 2 & -1 & 1 \end{bmatrix}.$$

Both have rank 2 (since A has full column rank 2). Only $A^T A$ is invertible, with

$$(A^T A)^{-1} = \begin{bmatrix} 1/5 & 0 \\ 0 & 1/6 \end{bmatrix}.$$

3. We start from

$$(AB)A = A.$$

Taking transpose, (keep in mind that AB is symmetric)

$$((AB)A)^T = A^T \Rightarrow A^T(AB) = A^T.$$

Hence

$$(A^T A)B = A^T.$$

Since $A^T A$ is invertible,

$$\boxed{B = (A^T A)^{-1} A^T}.$$

For this specific A ,

$$B = \begin{bmatrix} 1/5 & 2/5 & 0 \\ 1/3 & -1/6 & 1/6 \end{bmatrix}.$$

4. We have $(AB)^2 = (ABA)B = AB$. (Also because it is the orthogonal projection to $\text{Ran}(A)$.) So

$$p(AB) = (AB)^4 - 3(AB)^2 + 2(AB) + I = AB - 3AB + 2AB + I = I.$$

FYI, we actually should have

$$AB = \frac{1}{30} \begin{bmatrix} 26 & 2 & 10 \\ 2 & 29 & -5 \\ 10 & -5 & 5 \end{bmatrix}.$$

3. (Electrical Circuit)

This model arises from a simple electrical circuit, but all questions below are purely algebraic. We consider the block matrix

$$M = \begin{bmatrix} A^T & O \\ R & A \end{bmatrix},$$

where A is 3×2 and has rank 2, and R is invertible.

- (a) (3 points) Find a block row-swap matrix P that swaps the first block row and the second block row of M . List the size of every block in P .

- (b) (3 points) Let

$$\widetilde{M} = \begin{bmatrix} R & A \\ A^T & O \end{bmatrix}.$$

Recall that R is invertible, show that $\widetilde{M}$ admits a block LDU decomposition.

- (c) (3 points) Consider the system

$$\begin{bmatrix} A^T & O \\ R & A \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}.$$

Here $\mathbf{x}, \mathbf{b} \in \mathbb{R}^3, \mathbf{y} \in \mathbb{R}^2$. Solve for $\mathbf{x}, \mathbf{y}$ by expressing them in terms of $A, R, \mathbf{b}$. We assume that $A^T R^{-1} A$ is invertible.

- (d) (4 points) (HARD PROBLEM) Find a block matrix X such that

$$\begin{bmatrix} A^T & O \\ R & A \end{bmatrix} X = \begin{bmatrix} A^T R & A^T A \\ R^2 + A A^T & R A \end{bmatrix}.$$

Write X in block form. We assume that $A^T R^{-1} A$ is invertible.

Answer:

1.

$$P = \begin{bmatrix} O_{3 \times 2} & I_{3 \times 3} \\ I_{2 \times 2} & O_{2 \times 3} \end{bmatrix}, \quad PM = \begin{bmatrix} R & A \\ A^T & O \end{bmatrix}.$$

2.

$$\begin{bmatrix} R & A \\ A^T & O \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 \\ A^T R^{-1} & I \end{bmatrix}}_L \underbrace{\begin{bmatrix} R & 0 \\ 0 & -A^T R^{-1} A \end{bmatrix}}_D \underbrace{\begin{bmatrix} I & R^{-1} A \\ 0 & I \end{bmatrix}}_U.$$

3.

$$\begin{bmatrix} A^T & O \\ R & A \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}, \quad A^T R^{-1} A \text{ invertible.}$$

Method 1 (via $U^{-1}D^{-1}L^{-1}$): We swap to

$$\begin{bmatrix} R & A \\ A^T & O \end{bmatrix} \begin{bmatrix} \mathbf{y} \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}.$$

Then we can use the previous LDU result

$$\begin{aligned} L^{-1} \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix} &= \begin{bmatrix} \mathbf{b} \\ -A^T R^{-1} \mathbf{b} \end{bmatrix}, \\ D^{-1} \begin{bmatrix} \mathbf{b} \\ -A^T R^{-1} \mathbf{b} \end{bmatrix} &= \begin{bmatrix} R^{-1} \mathbf{b} \\ (A^T R^{-1} A)^{-1} A^T R^{-1} \mathbf{b} \end{bmatrix}, \\ U^{-1} &= \begin{bmatrix} I & -R^{-1} A \\ 0 & I \end{bmatrix}, \\ \Rightarrow \quad \mathbf{x} &= R^{-1} (I - A (A^T R^{-1} A)^{-1} A^T R^{-1}) \mathbf{b}, \\ \mathbf{y} &= (A^T R^{-1} A)^{-1} A^T R^{-1} \mathbf{b}. \end{aligned}$$

Method 2:

Since $A^T \mathbf{x} = \mathbf{0}$, we multiply $A^T R^{-1}$ on the left of $R\mathbf{x} + A\mathbf{y} = \mathbf{b}$, and we have

$$A^T \mathbf{x} + A^T R^{-1} A \mathbf{y} = A^T R^{-1} \mathbf{b}.$$

This gives

$$A^T R^{-1} A \mathbf{y} = A^T R^{-1} \mathbf{b}.$$

So we have

$$\mathbf{y} = (A^T R^{-1} A)^{-1} A^T R^{-1} \mathbf{b}.$$

And we have

$$\mathbf{x} = R^{-1}(\mathbf{b} - A\mathbf{y}) = R^{-1}(I - A(A^T R^{-1} A)^{-1} A^T R^{-1})\mathbf{b}.$$

4. We can reverse engineer this problem. We have

$$\begin{aligned} \begin{bmatrix} A^T R & A^T A \\ R^2 + AA^T & RA \end{bmatrix} &= \begin{bmatrix} A^T R & O \\ R^2 & O \end{bmatrix} + \begin{bmatrix} O & A^T A \\ O & RA \end{bmatrix} + \begin{bmatrix} O & O \\ AA^T & O \end{bmatrix} \\ &= \begin{bmatrix} A^T & 0 \\ R & A \end{bmatrix} \left(\begin{bmatrix} R & O \\ O & O \end{bmatrix} + \begin{bmatrix} O & A \\ O & O \end{bmatrix} + \begin{bmatrix} O & O \\ A^T & O \end{bmatrix} \right) \\ &= \begin{bmatrix} A^T & 0 \\ R & A \end{bmatrix} \begin{bmatrix} R & A \\ A^T & 0 \end{bmatrix}. \end{aligned}$$

Alternatively, as a thoughtless brute force method, just do block Gaussian elimination. It's not too bad. We start with

$$\begin{aligned} &\begin{bmatrix} A^T & O & A^T R & A^T A \\ R & A & R^2 + AA^T & RA \end{bmatrix} \rightarrow \begin{bmatrix} R & A & R^2 + AA^T & RA \\ A^T & O & A^T R & A^T A \end{bmatrix} \\ \rightarrow &\begin{bmatrix} R & A & R^2 + AA^T & RA \\ O & -A^T R^{-1} A & -A^T R^{-1} AA^T & O \end{bmatrix} \rightarrow \begin{bmatrix} I & R^{-1} A & R + R^{-1} AA^T & A \\ O & I & A^T & O \end{bmatrix} \\ \rightarrow &\begin{bmatrix} I & O & R & A \\ O & I & A^T & O \end{bmatrix}. \end{aligned}$$

In the RREF, the right half is the answer.

4. (Parabolic Shearing)

Let

$$V = \left\{ \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} : w = 0, x + y + z = 0 \right\}, \quad W = \left\{ \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} : z = 0, w + x + y = 0 \right\}.$$

- (a) (3 points) Find a basis for $V \cap W$, then extend it to a basis for V , and further extend to a basis for $V + W$.
- (b) (2 points) Let B be any 2×4 matrix such that $\ker(B) = V + W$. Compute the rank of the block matrix

$$\begin{bmatrix} B & 2B \\ 3B & 4B \end{bmatrix}.$$

- (c) (2 points) Suppose a linear transformation $A : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ satisfies $\ker(A - I) = V$ and $\text{Ran}(A - I) = W$, and in addition that

$$(A - I)e_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad (A - I)e_4 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Determine the unique matrix A consistent with all the above information.

- (d) (1 point) We use the matrix A in the last subproblem. Consider the curve (a parabola in $\mathbb{R}^4$)

$$\gamma(t) = \begin{bmatrix} 1 + t \\ -t - \frac{t(t-1)}{2} \\ \frac{t(t-1)}{2} \\ 0 \end{bmatrix}.$$

Show that points on this curve will remain on this curve after the transformation A , and determine s as a function of t such that

$$A(\gamma(t)) = \gamma(s).$$

Answer:

1.

$$V \cap W = \left\{ \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} : w = z = 0, x + y = 0 \right\} = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \right)$$

Then we have

$$V = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right),$$

$$V + W = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \right).$$

Note that the later two is not unique. The new vector for the basis of V could be anything with last three coordinates adding to zero, and not parallel to

$$\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

The new vector for the basis of $V + W$ could be anything with all coordinates adding to zero, and the first coordinate is not zero.

2. Note that for $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, we must have

$$B = \begin{bmatrix} s\mathbf{w}^T \\ t\mathbf{w}^T \end{bmatrix},$$

where s, t are not both zero.

Then

$$\begin{bmatrix} B & 2B \\ 3B & 4B \end{bmatrix} = \begin{bmatrix} s\mathbf{w}^T & 2s\mathbf{w}^T \\ t\mathbf{w}^T & 2t\mathbf{w}^T \\ 3s\mathbf{w}^T & 4s\mathbf{w}^T \\ 3t\mathbf{w}^T & 4t\mathbf{w}^T \end{bmatrix}.$$

Since the first two rows are parallel and the last two rows are parallel, the rank is at most two. Since s, t are not both zero, the rank is exactly two.

3. Set $X = A - I$. The information $X\mathbf{e}_1, X\mathbf{e}_4$ gives the first and fourth column of X . Let us work out the other two columns.

Since $\mathbf{e}_2 - \mathbf{e}_4, \mathbf{e}_3 - \mathbf{e}_4 \in V$, we have $X(\mathbf{e}_2 - \mathbf{e}_4) = X(\mathbf{e}_3 - \mathbf{e}_4) = \mathbf{0}$. So $X\mathbf{e}_2 = X\mathbf{e}_3 = X\mathbf{e}_4$. So we have

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then

$$A = I + X = \begin{bmatrix} 2 & 1 & 1 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4. Direct computation shows $A\gamma(t) = \gamma(t+1)$.

For a more in depth analysis, note that

$$\gamma(t) = \mathbf{e}_1 + tX\mathbf{e}_1 + \frac{t(t-1)}{2}X^2\mathbf{e}_1.$$

So we have

$$\begin{aligned} A\gamma(t) &= (I + X)\gamma(t) \\ &= \mathbf{e}_1 + X\mathbf{e}_1 + tX\mathbf{e}_1 + tX^2\mathbf{e}_1 + \frac{t(t-1)}{2}X^2\mathbf{e}_1 + \frac{t(t-1)}{2}X^3\mathbf{e}_1 \\ &= \mathbf{e}_1 + (t+1)X\mathbf{e}_1 + \left(\frac{t(t-1)}{2} + t\right)X^2\mathbf{e}_1 + \mathbf{0} \\ &= \mathbf{e}_1 + (t+1)X\mathbf{e}_1 + \left(\frac{(t+1)t}{2}\right)X^2\mathbf{e}_1 \\ &= \gamma(t+1). \end{aligned}$$

Here note that $X^3 = 0$.

5. (Abstract Linear Map on a Matrix Space)

Let $V = M_2(\mathbb{R})$ be the space of all real 2×2 matrices. Fix

$$A = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}, \quad T : V \rightarrow V, \quad T(X) = AX + XA.$$

Let

$$\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22}),$$

where

$$E_{ij} = \mathbf{e}_i \mathbf{e}_j^T.$$

Define another basis

$$\mathcal{C} = (C_1, C_2, C_3, C_4),$$

where,

$$C_1 = \begin{bmatrix} 4 & -1 \\ 4 & 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}, \quad C_3 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad C_4 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

- (a) (3 points) Find the change-of-coordinate matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ (so that $(X)_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}(X)_{\mathcal{B}}$), and its inverse $P_{\mathcal{B} \leftarrow \mathcal{C}}$.
- (b) (2 points) Find the matrix representation of T when the **domain** is expressed under basis $\mathcal{B}$ and the **codomain** under basis $\mathcal{C}$, i.e. $T_{\mathcal{C} \leftarrow \mathcal{B}}$.
- (c) (3 points) Find the matrix representation of T when both the domain and codomain are expressed under the same basis $\mathcal{B}$, i.e. $T_{\mathcal{B} \leftarrow \mathcal{B}}$. How is $T_{\mathcal{B} \leftarrow \mathcal{B}}$ and $T_{\mathcal{C} \leftarrow \mathcal{B}}$ related and through which change of coordinate matrix?
- (d) (1 point) Find a matrix $X \in V$ such that $T^2(X) \neq O$.
- (e) (1 point) Prove that whenever $A^2 = 0$, the associated operator $T(X) = AX + XA$ always satisfies $T^3 = O$.

Answer:

1. Each C_j written in the standard basis $\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22})$:

$$(C_1)_{\mathcal{B}} = \begin{bmatrix} 4 \\ -1 \\ 4 \\ 0 \end{bmatrix}, \quad (C_2)_{\mathcal{B}} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 4 \end{bmatrix}, \quad (C_3)_{\mathcal{B}} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad (C_4)_{\mathcal{B}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

Hence

$$P_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 4 & 4 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 4 & 0 & 0 & 1 \\ 0 & 4 & 0 & 0 \end{bmatrix}, \quad P_{\mathcal{C} \leftarrow \mathcal{B}} = (P_{\mathcal{B} \leftarrow \mathcal{C}})^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 & -\frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{4} \\ \frac{1}{4} & 1 & 0 & -\frac{1}{4} \\ -1 & 0 & 1 & 1 \end{bmatrix}.$$

2. Compute $T(E_{ij})$ and express each as a linear combination of C_1 and C_2 :

$$T(E_{11}) = C_1, \quad T(E_{12}) = C_2, \quad T(E_{21}) = -\frac{1}{4}C_2, \quad T(E_{22}) = C_1 - C_2.$$

Thus

$$T_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -\frac{1}{4} & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

3. Converting the codomain to basis $\mathcal{B}$:

$$T_{\mathcal{B} \leftarrow \mathcal{B}} = P_{\mathcal{B} \leftarrow \mathcal{C}} T_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 4 & 4 & -1 & 0 \\ -1 & 0 & 0 & -1 \\ 4 & 0 & 0 & 4 \\ 0 & 4 & -1 & -4 \end{bmatrix}.$$

4. Since $A^2 = 0$, we have $T^2(X) = 2AXA$. For $X = E_{11}$,

$$T^2(E_{11}) = 2AE_{11}A = 2 \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix} = 2 \begin{bmatrix} 4 & -2 \\ 8 & -4 \end{bmatrix} \neq 0.$$

The $(1, 1)$ entry equals 8, so E_{11} works.

5. In general,

$$T^3(X) = A^3X + 3A^2XA + 3AXA^2 + XA^3.$$

If $A^2 = 0$, all terms vanish, so $T^3(X) = 0$ for all $X \in V$.